Optimization for Deep Learning (OPT4DL)

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Advanced Training Methods: Adaptivity



- Recall we showed minibatch SGD satisfies

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{z_{\mathbf{1}:T}}\|\nabla f(w_t)\|^2 \lesssim \frac{\Delta}{T\eta} + \frac{L\eta\sigma^2}{2B} \tag{1}$$

  and we set the $\eta = \min\{\frac{1}{L}, \sqrt{\frac{2\Delta B}{L\sigma^2 T}}\}$ to get the best bound. But there are lots of unknowns...

- Here we focus on estimating variance on the fly

$$\mathbb{E}\|g_t - \nabla f(w_t)\|^2 \leq \sigma^2 \tag{2}$$

  from the stochastic gradients we have observed so far: $g_1, \ldots, g_t$. Note, these are NOT independent random quantities.

Towards Variance Estimation



- To gain an intuition, let us assume they were i.i.d. with zero means

$$\sigma^2 \geq \mathrm{Var} = \mathbb{E}\|g\|^2 \approx \frac{1}{t}\sum_{\tau=1}^{t} \|g_\tau\|^2 \tag{3}$$

such that $t\sigma^2 \approx \sum_{\tau=1}^{t} \|g_\tau\|^2$.

- Let us then use this expression in our learning rate formula and SGD:

$$w_{t+1} = w_t - \eta_t g_t, \qquad \eta_t = \frac{C}{\sqrt{\sum_{\tau=1}^{t} \|g_\tau\|^2 + \kappa^2}} \tag{4}$$

for some constant C where we also added $\kappa > 0$ for numerical stability.

- This method is called adaptive SGD.
- Like Normalization methods, we are normalizing a key quantity



- Note that $\eta_t = \frac{C}{\sqrt{\sum_{\tau=1}^{t} \|g_\tau\|^2 + \kappa^2}}$ is not deterministic anymore as it depends on the random gradients.
- For implication, consider the part of proof from SGD:

$$\begin{aligned} f(w_{t+1}) &\leq f(w_t) + \langle w_{t+1} - w_t, \nabla f(w_t) \rangle + \frac{L}{2}\|w_{t+1} - w_t\|^2 \\ &= f(w_t) - \eta_t \langle g_t, \nabla f(w_t) \rangle + \frac{L\eta_t^2}{2}\|g_t\|^2 \end{aligned} \tag{5}$$

where conditioned on w_t (equivalently $z_1, \ldots, z_{t-1}$)

$$\mathbb{E}_{z_t}[\eta_t \langle g_t, \nabla f(w_t) \rangle | w_t] \neq \eta_t \mathbb{E}_{z_t}[\langle g_t, \nabla f(w_t) \rangle | w_t] \tag{6}$$

as η_t is depends on g_t

Handling Randomness of η_t

- Let $\eta_t = \frac{C}{\sqrt{\sum_{\tau=1}^{t} \|g_\tau\|^2 + \kappa^2}}$ for $t \geq 1$ and $\eta_0 = C/\kappa$
- Note that our learning rate schedule is causal,meaning it does not depend on future stochastic gradients.
- Furthermore, our schedule is non-increasing as we are accumulating non-negative terms $\|g_\tau\|^2$ in its denominator.
- We can leverage these properties

$$
\begin{aligned}
f(w_{t+1}) &\leq f(w_t) + \langle w_{t+1} - w_t, \nabla f(w_t) \rangle + \frac{L}{2}\|w_{t+1} - w_t\|^2 \\
&= f(w_t) - \eta_t \langle g_t, \nabla f(w_t) \rangle + \frac{L\eta_t^2}{2}\|g_t\|^2 \\
&= f(w_t) - \underbrace{\eta_{t-1}\langle g_t, \nabla f(w_t) \rangle}_{\text{Term 1}} + \frac{L\eta_t^2}{2}\|g_t\|^2 + \underbrace{(\eta_{t-1} - \eta_t)\langle g_t, \nabla f(w_t) \rangle}_{\text{Term 2}}
\end{aligned}
\tag{7}
$$

Handling Randomness of η_t (cont'd)

$$f(w_{t+1}) = f(w_t) - \underbrace{\eta_{t-1}\langle g_t, \nabla f(w_t)\rangle}_{\text{Term 1}} + \frac{L\eta_t^2}{2}\|g_t\|^2 + \underbrace{(\eta_{t-1} - \eta_t)\langle g_t, \nabla f(w_t)\rangle}_{\text{Term 2}} \tag{8}$$

- Term 1 has no issue with randomness anymore by causality
- Term 2 can be uniformly bounded by $(\eta_{t-1} - \eta_t)G^2$ assuming uniformly bounded (stochastic) gradients, e.g., by imposing G-Lipschitzness of sample loss $\ell(w, z)$
- Using our previous proof techniques, notably the telescoping term $\sum_{t=1}^{T}(\eta_{t-1} - \eta_t)G^2 = G^2(\eta_0 - \eta_T) \leq G^2\eta_0 = G^2C\kappa^{-1}$, we can establish the following result

Theorem

Assume $f(w) = \mathbb{E}[\ell(w, z)]$ is L-smooth and $\ell(w, z)$ is G-Lipschitz. Furthermore, assume we have access to an SFO and that $z_1, \ldots, z_T$ are statistically independent. Then, adaptive SGD with learning rate $\eta_t = \frac{C}{\sqrt{\sum_{\tau=1}^{t}\|g_\tau\|^2 + \kappa^2}}$ satisfies $\mathbb{E}\|\nabla f(\hat{w})\|^2 \simeq \mathcal{O}(\kappa^{-1}/\sqrt{T})$.

Integrating Exponential Moving Average (EMA)



- Our method is basically a simple version of famous methods such as Adagrad, Adadelta, RMSProp, and ADAM
- A key modification is using weighted averaging, putting emphasis on most recent observations

$$\eta_t = \frac{C}{\sqrt{\sum_{\tau=1}^{t} \|g_\tau\|^2 + \kappa^2}} \quad \rightarrow \quad \eta_t = \frac{C}{\sqrt{\sum_{\tau=1}^{t} \tilde{\beta}^{t-\tau}\|g_\tau\|^2 + \kappa^2}}, \quad 0 < \tilde{\beta} \leq 1$$

- In terms of moving averages, $\eta_t = \frac{C}{\sqrt{t \times S_t + \kappa^2}}$ where

$$S_t = (1-\frac{1}{t})S_{t-1} + \frac{1}{t}\|g_t\|^2 \quad \rightarrow \quad S_t = \beta S_{t-1} + (1-\beta)\|g_t\|^2 \text{ (exponential moving average (EMA))}$$

- For large t, simple average diminishes the contribution of new observation, but with a fixed β, the contribution of new observation is not diminished with time.
- $1 - \frac{1}{t} \rightarrow \beta$ Gives an idea that in practice β should be close to 1, e.g., $\beta = 0.99$

EMA and Training Instability



- One potential issue of EMA is losing the non-increasing property of η_t

$$\eta_t \leq \eta_{t-1} \equiv t \times S_t \geq (t-1) \times S_{t-1} \tag{9}$$

- The condition $t \times S_t \geq (t-1) \times S_{t-1}$ always happen for simple averaging, but if

$$\|g_t\|^2 \leq S_{t-1}\Big(1 - \frac{1}{t(1-\beta)}\Big) \tag{10}$$

  it won't happen for EMA. Likely to occur when t is large or β is small
- As a result, learning rates start to go up and lead to training instability and divergence issues (even for convex problems).
- Fix 1: Ensure $\beta \approx 1$ and do not overtrain
- Fix 2: Implement safeguarding: $S_t \leftarrow \max(S_t, S_{t-1}\frac{(t-1)}{t})$ to ensure the non-increasing property of η_t holds (An idea leading to AMSgrad)

Per-coordinate learning rates and curvature adaptivity



- In deep learning, different parameters may exhibit gradients with widely varying magnitudes.
- The multi-dimensional loss function in deep learning could mean some directions exhibit steep curvature while others are relatively flat.
- Per-coordinate learning rates: each parameter or group of parameters is updated with a learning rate adapted to its own gradient statistics.

$$w_{t+1}(i) = w_t(i) - \eta_t(i) g_t(i), \qquad \eta_t(i) = \frac{C}{\sqrt{\sum_{\tau=1}^{t} |g_\tau(i)|^2 + \kappa^2}} \tag{11}$$

- Similar motivation as normalization methods
- Can be analyzed by consider d one dimensional problems as

$$\langle g_t, w_t - w^* \rangle = \sum_{i=1}^{d} g_t(i)[w_t(i) - w^*(i)] \tag{12}$$

(And similar behavior for other important terms)